

Sukhad Tomar

Delhi | sukhadtomar@gmail.com | +91 9219855071 | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building AI and machine learning applications using Python, LLMs, RAG, and agentic workflows. Skilled in LangChain, LangGraph, vector search, and end-to-end ML pipelines, with experience designing and developing practical AI solutions across different use cases. Strong understanding of LLM-based systems, retrieval pipelines, workflow automation, and machine learning, with a focus on building reliable and scalable applications.

SKILLS

- **Programming:** Python, SQL
- **Generative AI:** LLMs, Generative AI, RAG, Prompt Engineering, LangChain, LangGraph, Agentic AI, Function Calling, Structured Outputs, MCP
- **AI / ML:** Machine Learning, Deep Learning, NLP, Scikit-learn, TensorFlow, Model Evaluation, LLM Evaluation
- **Retrieval & Data:** Embeddings, Semantic Search, Vector Databases, Vector Search, Hybrid Search, Reranking
- **Backend & APIs:** FastAPI, REST APIs, Pydantic
- **Automation & Engineering:** n8n, Multi-Agent Systems, Workflow Orchestration, Docker, Git, CI/CD
- **Soft Skills:** Communication, Problem Solving, Team Collaboration

EXPERIENCE

AI Engineer | QuickReturns Sept 2025 – Present

- Developed an LLM-based return-reason classification pipeline in Python that converts unstructured customer return requests into structured categories, feeding analytics dashboards and downstream automation.
- Implemented a RAG-based policy retrieval pipeline that ingests merchant return and exchange policies and retrieves relevant policy context for each incoming request to support downstream decision workflows.
- Designed a LangGraph-based agentic workflow that combines retrieved policy context with LLM reasoning to determine the appropriate action for return or exchange requests, including approval, rejection, or escalation.
- Worked on a product recommendation pipeline for ConvertX using embedding-based similarity search to surface relevant exchange alternatives during the return flow, helping convert refunds into exchanges.

PROJECTS

Data Science Copilot Python | LangGraph | FastAPI | Scikit-learn | SHAP

- Built an AI-powered Data Science Copilot that automates tabular ML workflows, from natural-language objectives and data profiling to model training, evaluation, explainability, and business recommendations.
- Designed a LangGraph-based multi-agent system with an LLM-driven Planner that creates and validates dataset-specific ML plans and routes the workflow based on the required steps.
- Built a self-healing ML pipeline that detects failures, identifies the cause using heuristics with LLM fallback, applies predefined fixes, and retries failed steps while keeping LLM reasoning separate from deterministic Python execution.

DocIntel AI Python | FastAPI | LangChain | FAISS | BM25 | HuggingFace

- Built a Hybrid RAG-based PDF question-answering system that combines document ingestion, chunking, embeddings, FAISS, BM25, and LLMs to answer questions from uploaded documents.
- Combined BM25 keyword search with FAISS semantic search and used Reciprocal Rank Fusion (RRF) to find and rank relevant document chunks for each query.
- Added a routing layer that looks at semantic relevance, keyword matches, and agreement between the retrievers to decide whether to use document context or the general LLM.
- Added an agentic retrieval loop that checks the retrieved evidence, rewrites the query when needed, and retries retrieval within a fixed limit, with PDF page references included in document-based answers.

CERTIFICATIONS

Machine Learning, CodeBasics (Jun 2025) | Python, CodeBasics (Feb 2025)

EDUCATION

HR Institute of Technology, Ghaziabad, UP, B.Tech, CSE (Jun 2022 – Jun 2026)